A Minimal Epistemic Core

Differentiation, Receptive Dependence, and Registration

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ABSTRACT. Theories of knowledge usually start from a subject who perceives or believes. This paper asks what must already hold for anything to appear at all, and proposes a minimal answer in two theses. The first is that what appears of a thing is nothing over and above differences that it bears. The second is that a difference appears only where something is as it is in virtue of it. On this basis the paper defines registration, the holding of a condition in virtue of a difference between actual terms, and proposes it as the elementary form of appearing. Registration requires no alternatives, no resemblance or pairing between what registers and what is registered, no order of occurrences, and no knowing subject; a single dependent condition suffices on the receiving side. Two cases, one causal and one constitutive, illustrate the structure. Since nothing is perceived that does not appear, the two theses are offered as necessary conditions of perception and of the knowledge that rests on it. What a subject adds is left to further work.

1. Introduction

Theories of knowledge ordinarily begin with a subject who believes, judges, or is justified, and ask under what conditions such states amount to knowledge. This paper asks what must hold before that: what is required for anything to appear at all.

Its answer is a proposal in two parts. The first is that what appears of a thing is nothing over and above differences that it bears, since nothing appears where everything is alike. The second is that a difference appears just where something is as it is in virtue of it. The paper calls the structure formed by these two parts *registration*.

To appear, in this paper, is to show up. The word is not used in the sense of seeming, as opposed to being, and it implies no subject to whom something appears. Perception and knowledge involve a subject and are not treated here; but since nothing is perceived that does not appear, the conditions proposed for appearing are, on this account, necessary for them as well. The conditions are formal. They are stated without reference to space, time, matter, or causation,

and the examples drawn from particular domains show how the conditions are met there without fixing what they are.

The exposition uses ordinary logical resources—identity, distinction, relation, and inference—to state its commitments. This does not attribute those resources to what the commitments describe, and in particular not to a knowing subject. A distinction drawn in the exposition is not thereby a distinction drawn by its object.

The proposal has close neighbours. Dretske (1981) analyses the information carried by a signal in terms of nomic dependence and conditional probability, and separates it from meaning and belief. Floridi's diaphoric definition of data takes a lack of uniformity—a difference—as the most elementary constituent of information (Floridi 2011), and his informational structural realism holds that the relation of difference is constitutive of its relata (Floridi 2008). Smith's metaphysics of presence makes *registration* its central notion, as the way in which a world comes to be taken as one of objects (Smith 1996).[1] Enactivist accounts describe cognition as the structural coupling of a system with a medium, both of which belong to a larger organization (Maturana and Varela 1987; Thompson 2007). The present account differs from these in three respects: what is depended on is always a difference between actual terms; the dependence is stated without probabilities, causes, or alternatives; and its direction is kept distinct from any order of occurrences.

Sections 2 and 3 concern the difference borne by what is under consideration, and Sections 4 and 5 the dependence and the receiving side. Section 6 defines registration, Section 7 works through two cases, and Section 8 separates the direction of the dependence from any order.

2. Differentiation

Let F designate what is under consideration. Calling it a *field* fixes the scope of the inquiry; it implies neither a spatial region, nor a collection of discrete objects, nor something already presented to a subject. Nor does the notation establish a boundary intrinsic to reality: F identifies what the account concerns, and any claim about its actual boundaries would require separate justification.

Take anything that is not alike throughout. Something in it is one way and something else in it is another: there are two, and in some respect they are not alike. That is a difference. A difference between two actual terms, in a given respect, is a *contrast*. The two that differ are its *terms*, that in which they differ is its *respect*, and whatever has the two terms *bears* the contrast. Where more than two terms differ, they are taken two at a time. Two contrasts are the same

1. The present use of the term is narrower. Smith's registration is performed by subjects and yields objects; registration here presupposes neither.

when their terms and their respect are the same, so that the same two terms may bear two contrasts, in two respects. A contrast belongs to what is under consideration, not to the account that speaks of it; how an account speaks of a contrast, by means of an articulation, is the subject of Section 3.

(D) DIFFERENTIATION

What appears of F is nothing over and above contrasts that F bears.

To appear is to stand out from something: where everything is alike, nothing stands out. This holds even when F is taken whole: what appears is then its difference from something beside it, and what is under consideration is the two together. If F is everything, there is nothing beside it, and what appears of it is the differences within it. (D) concerns what appears of F, not what F is. Whether a thing has a nature beyond its differences is left open; what (D) claims is that no such nature appears apart from them, so that what enters no contrast does not appear.

Both terms of a contrast are actual, "actual" being meant without tense. A difference between something actual and an alternative, that is, a way that nothing actual is, is not a contrast. An alternative would have to be somewhere, in a stock of possibilities, in a repertoire belonging to whatever receives, or in a description, or else be a modal fact about the thing itself; and none of these is assumed here. Three things therefore do not appear in the way described in this paper: which way something is as against a way that nothing is, anything of a term that differs from nothing actual, and that two terms are alike.

To say that F bears a contrast is to say that there is a difference in F, not that F could be differentiated. It does not say which contrasts F bears, whether one differentiation of F is privileged, or whether what differs are objects, qualities, regions, or values. It implies nothing about change or order, and it involves no knowing subject, since that F bears a contrast does not imply that anything differentiates it.

The contrasts concerned are those of F itself: the account concerns a differentiation attributable to its target, rather than one supplied entirely by an unrelated description. A difference that holds, even in part, because the account has specified it belongs to the description and is left out. This does not exclude the respects in which the account chooses to speak of F: a respect that the account chooses is not thereby one that it supplies, and what matters is whether the terms differ in it apart from the choosing.

3. Articulation

An account speaks of a particular contrast of F by means of an *articulation*, written

$$\langle a, \delta, b \rangle_F,$$

which names two terms, a and b, and a respect, δ. The articulation is *specified* by the account, not by F, and it is *realized* when a and b are actual terms of F that differ in δ. Specification selects a contrast and does not create one. It can therefore fail: a name may pick out nothing, or the terms may not differ in the respect named. The order in which the terms are written, their labels, and their grouping belong to the articulation, not to F. Specification fixes what an inquiry concerns, as F fixes its scope, but it does not supply what the inquiry finds.

The terms of a realized articulation are both actual: they are not alternatives of which one is the case and the other not. Nor does the schema assume that a and b exist before the contrast as independent objects.[2]

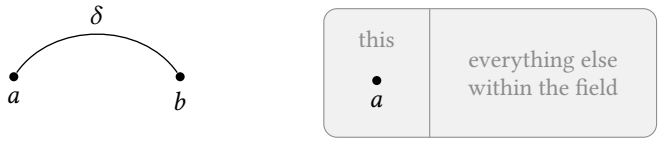
Two misreadings of the schema should be excluded. The first takes δ to be a third item situated between a and b. To treat it as a separator would presuppose spatial organization, whereas δ is the respect in which the two differ. The second takes b to comprise everything that remains once a has been selected. That reading would require an exhaustive partition of F, and a contrast between two terms is not equivalent to a division into "this" and "everything else within the field": the latter adds the requirement that every case within the scope fall on one side or the other (Figure 1).

A realized articulation remains an articulation of F, and nothing said so far implies that its contrast appears anywhere. What that requires is the subject of the next section.

4. Receptive Dependence

That F bears a contrast and that something else is in some condition are two facts, and nothing said so far connects them; they could be entirely unrelated.

2. That a difference need not be grounded in terms identified before it is argued by Deleuze in his critique of the subordination of difference to identity (Deleuze 1994, ch. 1); the present account leaves the question open. Deleuze also rejects the possible as a mere image of the real, and the present account likewise uses no stock of possibilities.



(a) contrast (b) exhaustive partition

FIGURE 1. Two constructions over F. (a) An articulation $\langle a, \delta, b\rangle_F$: a and b mark terms, both actual, and δ, drawn as a relation rather than as an item between them, is the respect in which they differ. (b) An exhaustive partition, which adds the requirement that every case fall on one side or the other. Only (b) is drawn as a region, because only (b) carries the extensional commitment that a region suggests; in (a) the placement of the marks has no spatial meaning.

For a contrast of F to appear, a further commitment is needed.

(C) RECEPTIVE DEPENDENCE

A contrast of F appears only where some condition of something holds, at least partly, in virtue of it.

A *condition* is something's being a certain way. (C) takes (D) for granted, that what appears is a contrast, and adds the dependence.

Two things may both be the case without either having anything to do with the other, or one may be the case because the other is. When one thing is the reason why another is so, the second holds *in virtue of* the first. (C) says that appearing is of this kind, and it is in this sense receptive: where a contrast of F appears, something is as it is because of it. The relation is called *dependence* for short. In what follows, K is a contrast of F and q a condition that holds in virtue of it.

Three things hold of dependence. It relates only what holds, so that K and q both hold. Nothing holds in virtue of itself. And it has a direction: that q holds in virtue of K does not bring with it that K holds in virtue of q.[3] Whether it is also transitive is left open.

The dependence is on the difference itself, not merely on how one of its terms is: a condition that held in virtue of how a is alone would not depend on the contrast of a and b. This does not oppose the contrast to what the two terms are together. What the condition depends on is what they are together, insofar as they differ.

Dependence is neither resemblance nor pairing. Something elsewhere may be like the contrast, and its parts may be paired with the terms of the contrast,

3. The first and third features are familiar from the literature on metaphysical ground (Fine 2012). Dependence is not thereby identified with ground.

without holding in virtue of it (Figure 2).[4] Both relations can be read in either direction, and neither ties one thing to this contrast rather than to any other that is like it. Nor does agreement show dependence: where two things agree because both depend on a third, their agreement does not establish that either depends on the other. (C) requires none of these relations, and in particular no pairing.

Nor is dependence the same as what would have held otherwise. A condition q may hold in virtue of K although, had K not held, something else would have brought q about; and it can be that neither K nor q holds without the other although both depend on a third. What would have held otherwise is evidence of dependence, not what it consists in.[5]

The dependence may be partial: q may hold in virtue of K and of other things as well, among them how its bearer is.[6] In particular, a condition may depend on several contrasts. It then stands in the same relation to each, and none of them is singled out.

Bateson's definition of information as "a difference which makes a difference" is close to this proposal (Bateson 1972), and MacKay had earlier described information as a distinction that makes a difference (MacKay 1969). In the present terms, the difference is K, and its making a difference is q's holding in virtue of it. The dependence itself is what matters, and it should not be replaced by an appeal to "systematic" or "relevant" coupling, which brings in a standard of relevance or a regularity across cases.

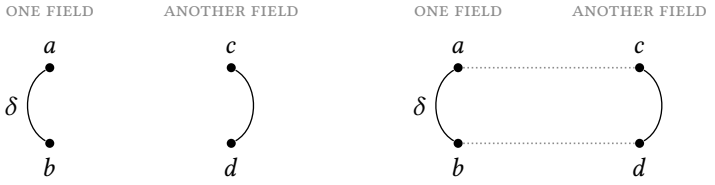
What the dependence consists in differs from domain to domain: it may be causal, logical, or constitutive. What is required is that there be such a dependence, not that it be of one kind.

The direction of the dependence gives two roles. In an instance of it, what bears the contrast is the *target*, and what bears the dependent condition is the *receiver*. The roles are places in the relation, not natures of what occupies them: the same thing may be target in one instance and receiver in another.

4. Goodman (1976, ch. 1) makes the parallel point about representation: resemblance is reflexive and symmetric, whereas representation is neither, so resemblance cannot be what representation consists in.

5. Counterfactual analyses of dependence require alternatives to the actual case. Where contrasts hold necessarily, they meet a further difficulty: on the standard semantics, counterfactuals with impossible antecedents are all vacuously true (Lewis 1973); for an alternative semantics, see Nolan (1997).

6. This accommodates the enactivist point that a perturbation triggers, but does not specify, the changes it occasions in a system, which are fixed by the system's own structure (Maturana and Varela 1987).



(a) resemblance (b) pairing

FIGURE 2. Two relations that are not dependence. (a) Resemblance: two contrasts are alike. (b) Pairing: a is paired with c, and b with d. The pairing is drawn without direction, since it can be read either way.

5. The Receiving Side

Let R designate what bears the dependent condition.[7] R need not be a person, a conscious subject, or an independently bounded system, and it may be an aspect of a larger organization that also includes F, since a distinction of roles does not establish two substances.

On the receiving side a single condition is enough. It is actual, but it need not be like the contrast of F in any way, and nothing in it is assigned to one term of the contrast rather than the other. Nor is a further contrast borne by R required: R need not be in a second condition from which the first differs.

The account attributes no act of distinguishing to R. That the condition of R holds in virtue of a contrast of F is a fact about the contrast and the condition, not something that R has to note; it is the account that picks them out.

Since the condition of R is not itself a contrast, it does not appear merely by holding. It can appear only as (D) allows: where it differs from something actual, and that contrast is in turn depended on.

Neither side comes with the other. A contrast of F may hold while nothing depends on it, and R may be in conditions that depend on nothing in F.

This fixes how much individuation is required. There must be enough to distinguish the contrast from the condition that depends on it. There need not be enough to identify enduring objects or successive states of a continuing subject. To assume that would build into the foundation a persisting, bounded bearer—which is what an account of a knowing subject must eventually explain.

7. The vocabulary of reception recalls Kant's characterization of sensibility as receptivity, the capacity to acquire representations through being affected by objects (Kant 1998, A19/B33, cf. A50/B74). The receiver introduced here is thinner: it involves no representations, and nothing is claimed about forms of intuition.

6. Registration

DEFINITION OF REGISTRATION

A contrast K is registered where a condition q of something holds, at least partly, in virtue of K. What bears q registers K.

Registration is always of a particular contrast, so that to say that R registers F without qualification is to leave the claim incomplete. It is an actual relation between a contrast and a condition that both hold, not a disposition of R to respond (Figure 3).

Registration is the elementary form of appearing: every registration is the appearing of one contrast. This is the proposal of the paper; it does not follow from (C), which gives only a necessary condition. Anything more that appears is, by (D), contrasts in some organization, with nothing further appearing besides them. The only organization available so far is the dependence itself, in two ways: one condition may hold in virtue of several contrasts, and a condition may enter a contrast that is registered in turn. That this is enough for the three things set aside in Section 2 is a hypothesis, which is not shown here.

The claim is broad. A contrast appears wherever it contributes to why some condition holds, whether or not the condition displays the contrast, distinguishes it from others, or keeps anything of it.

Registration is supplied by neither side alone. It involves three things: the contrast, the condition, and the dependence of the second on the first. Keeping them distinct prevents registration from being attributed to the target merely because the target is differentiated, or to the receiver merely because it is in some condition.

Registration involves no understanding, identification, or judgement on the part of the receiver. Where registration is realized physically, it is found in systems to which nobody would attribute knowledge, and by itself it does not place anything "in the logical space of reasons".[8] It requires no transfer from F to R, no resemblance, and no representational space or repertoire in R.

What is registered is always a difference between actual terms: that a differs from b. Accounts of information and knowledge that begin from alternatives presuppose a set among which the actual case is singled out;[9] registration

8. The phrase is from Sellars (1956, § 36); Brandom (1994, ch. 4) develops the contrast between such reliable differential responsiveness and conceptual capacities. In perception, Dretske's non-epistemic seeing, in which an object is visually differentiated from its surroundings without any belief about it, contains a registration in the present sense (Dretske 1969).

9. Shannon's theory measures information as selection from a set of possible messages (Shannon 1948), and Dretske's informational content is defined by conditional probabilities over such alternatives (Dretske 1981). In epistemology, relevant-alternatives theories and contrastivism analyse knowledge as the elimination of alternatives

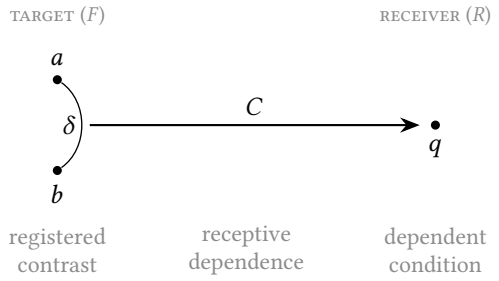


FIGURE 3. Registration. A condition q of R holds at least partly in virtue of the contrast of a and b in δ. The arrow runs from the contrast, not from either term, and denotes receptive dependence only: it represents neither a transfer nor an order of occurrences. The columns are roles, not regions.

involves no such set.

7. Two Cases

Two cases show the structure at work, one in which the dependence is causal and one in which it is constitutive.

In the first, let a and b be two bodies of unequal mass resting on the pans of a balance, and let the beam be inclined. The contrast K is that of the bodies in mass, and the condition q is the beam's being inclined. This condition holds, in part, in virtue of K: the beam is inclined because the two sides are unequally loaded, not because of the mass of either body considered alone. How far it is inclined depends on more, namely on how much the masses differ, on how the balance is built, and on where the bodies rest. The balance thus registers the difference in mass, and it does so without any repertoire of the masses that the bodies might have had.

In the second, let a and b be two parts of a whole W that differ in a respect δ. The contrast K is their difference in δ, and the condition q is the whole's being non-uniform in δ. Here q holds in virtue of K, since W is non-uniform because two of its parts differ. Nothing causes anything, and nothing passes from one thing to another: the whole registers a contrast within itself, through a condition of its own.

In neither case would anyone speak of perception or knowledge, since perception and knowledge involve a subject, and there is none here.

(Dretske 1970; Lewis 1996; Schaffer 2004); counterfactual and interventionist analyses of dependence share the same structure (Lewis 1973; Woodward 2003).

8. Registration and Order

Registration involves no order. The dependence has a direction, but the direction of a relation is not an ordering of occurrences: that q holds in virtue of K does not place either before the other.

For this reason a receiver cannot, on this basis alone, be said to *acquire* what it registers, to *retain* it, or to *compare* it with something else.[10] Retention requires that one receiver be identified across distinct places of some ordering. Acquisition requires that a place at which a contrast is not registered stand in an asymmetric relation to one at which it is. Neither follows from (C).

What the two require is at least an asymmetric relation among places, without any requirement that every two places be comparable, and without a metric, linearity, or a universal index. Such a relation may still run in a circle, so it is less than an order in the usual sense, and what more is needed is left open. Temporal succession is one relation of this kind. Whether a relation that is not temporal, such as the order of steps in a proof, could serve for acquisition or retention is also left open.

If receptive dependence is written with a directed symbol, such as an arrow from F to R, that symbol should not also serve, without further explanation, to represent order. The direction of dependence and the asymmetry of an ordering play different roles and should be kept notationally distinct.

9. Conclusion

Table 1 summarizes the elements of the account.

Taken together, these elements make a single proposal: what appears of a thing is nothing over and above contrasts that it bears, and a contrast appears just where something is as it is in virtue of it, with target and receiver distinguished as roles in that dependence.

Five questions remain open. The first is what the dependence consists in within a given domain, and whether it can be explained by more elementary relations; writing it as an arrow would not answer this. The second is what must be added to a registration for something to appear to a subject, that is, to be perceived. The third is how which way a term is, and that two terms are alike, can appear through contrasts alone. The fourth is what acquisition and retention require beyond an asymmetric relation. The fifth is whether what

10. The three operations closely parallel the synthesis of apprehension, reproduction, and recognition in the first edition of Kant's Transcendental Deduction (Kant 1998, A98–A110), and Husserl's distinction between retention and recollection (Husserl 1991) analyses the second of them. Both treat the operations as presupposing temporal form; the present account speaks only of order, of which temporal order is one kind.

TABLE 1. The elements of the account, with what each permits and what each leaves unestablished.

ELEMENT	WHAT IT PERMITS	WHAT IT DOES NOT ESTABLISH
(D) Differentiation	A difference attributable to what is under consideration, which is what appears of it.	Which contrasts F bears; that they appear anywhere; what F is beyond them.
Actual terms only	A core without possibilities or a repertoire.	Which way a term is as against an alternative; anything of a lone term; that two terms are alike.
Specified articulation	Terms and the respect in which they differ.	Its own realization, an exhaustive partition, or objecthood.
(C) Receptive dependence	Registration of a contrast; the roles of target and receiver.	Understanding, representation, order, acquisition, or retention.
The receiver R	A condition that depends on a contrast.	A knowing subject; a second condition; any resemblance to the contrast.

is said of things that cannot act on whoever speaks of them, such as things picked out by description, future events, and mathematical objects, rests on registration at all.

The account ends at registration. Order, memory, recognition, and representation have not been presupposed in reaching it, and none of them has yet been derived from it.

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